

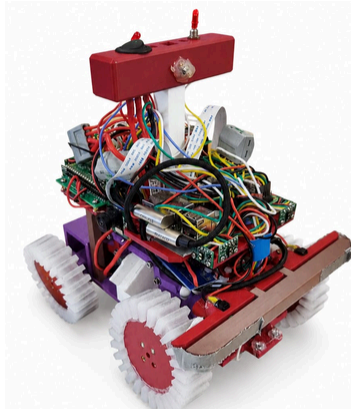


RRR Kabelmüsl

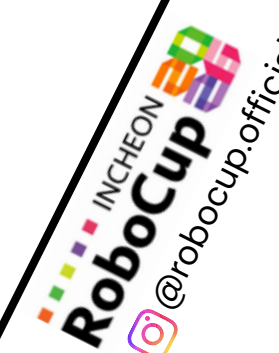
RoboCup Junior Rescue Maze

SFZ Tuttlingen, Germany

@kabelmuesli



Our Team



1 TEAM OVERVIEW

- Joel**
 - Team Captain
 - Software
 - Debugging
- Simon**
 - Software
 - Electronics
 - Marketing
- Felix**
 - Hardware
 - Electronics
 - Assembly
- David**
 - Hardware
 - Assembly
 - Design

2 ROBOT HARDWARE

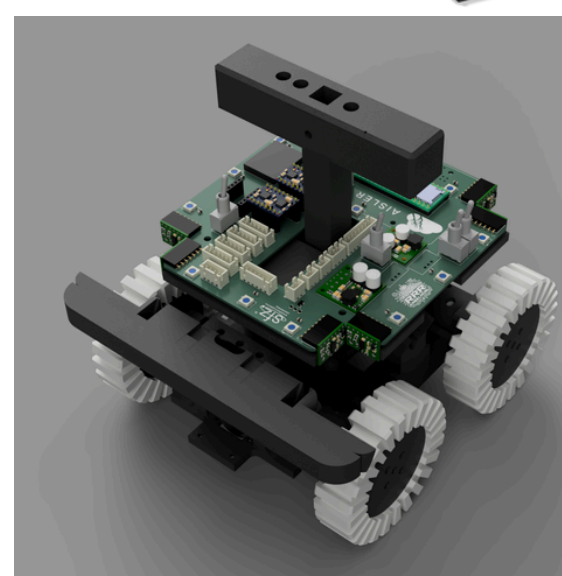
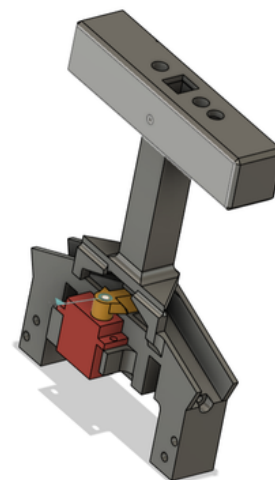
- Gravity-Fed Tower: Stores kits vertically
- Servo Escapement: Uses servo and two-armed catch
- Single-Kit Release: Drops the bottom kit while catching the rest

Sensors:

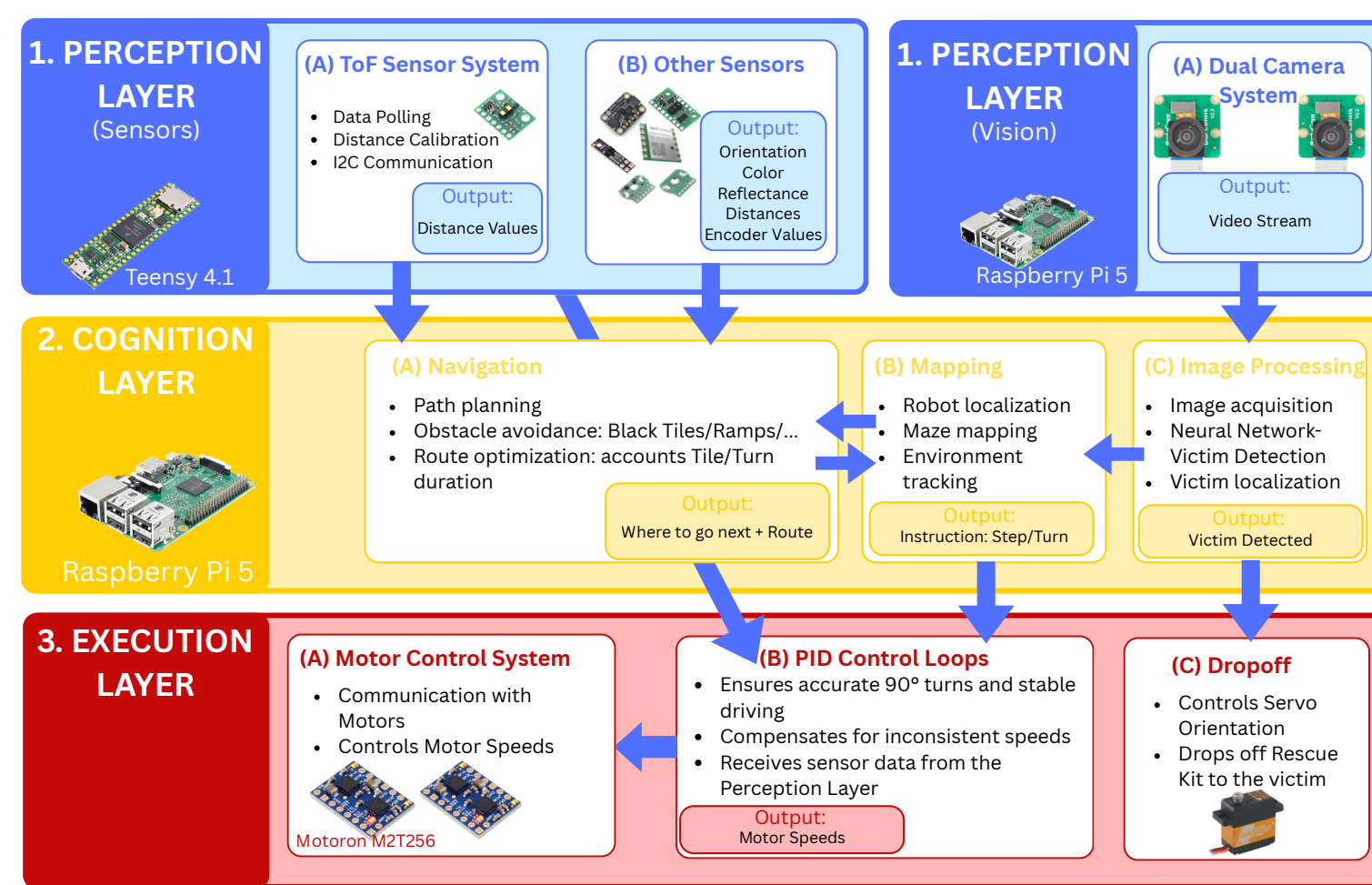
- 8x VL53L1X
- 3x VL53L8CX
- 1x QTRXL-HD-01A
- 1x APDS 9960
- 1x WT901

Motors:

- 4x Pololu Metal Gearmotor 20Dx55L



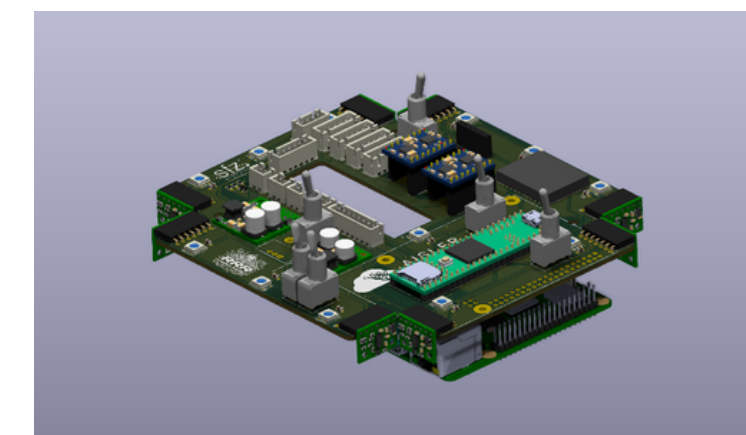
3 SOFTWARE ARCHITECTURE



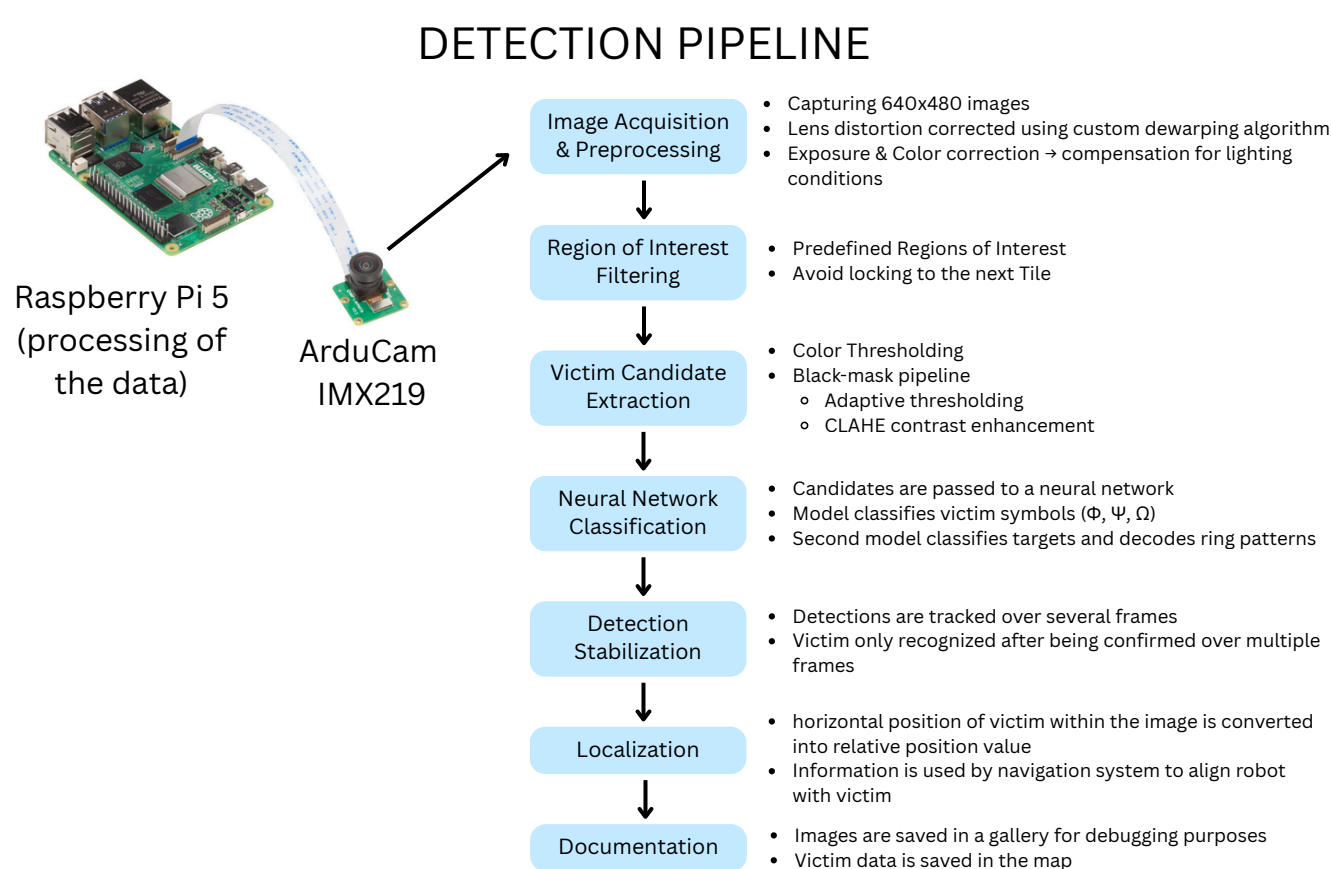
4 Electronics

CUSTOM PCB

- Designed to improve reliability and reduce wiring complexity
- Integrated 12 V, 5 V, and 3.3 V power supplies
- Onboard sensor integration
- Modular connectors for additional sensors
- Integrated motor drivers
- Compact, lightweight design
- Simplified assembly and troubleshooting



5 VICTIM DETECTION



6 MAPPING & NAVIGATION

- Localization & Mapping**

Distance sensors, wall measurements, and the gyro are used to estimate the robot's position and orientation.
- Path Planning**

Shortest-path algorithm determines optimal route to next unknown cell. Turn Costs are included so smoother paths are preferred
- Motion Control**

PID-Controller regulates movement and turning. Wall distances and heading errors are used for correction
- Obstacle Handling**

Black tiles, ramps, and unexpected obstacles are detected. Recovery behaviors allow the robot to continue operating after disturbances.
- Navigation**

The robot builds a grid map of the maze while exploring. A Dijkstra-based path planner uses this map to find the most efficient routes to the next unknown tile or back to the starting tile.

7 INNOVATION

Innovation: coupled axle suspension system with ball joints

- Each two wheels are rigidly mounted on one axle
- The front and rear axles are connected to each other via rods
- The connection uses ball joints
- This enables the axles to adapt to uneven ground and tilt
- Result: All four wheels remain in contact with the ground at the same time
- Advantage: high stability and consistent traction even on uneven terrain



8 ACCOMPLISHMENTS

2024 (Maze Entry)

1st German Open
3rd European Championship

2025 (Maze)

4th German Open
5th European Championship

2026 (Maze)

1st German Open

9 Our Partners



Thank you for making this possible!

10 CONTACT

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